Infosys Virtual internship

Week-1			DATE	14 Feb 2026
Dept./Division	ARTIFICIAL INTELLIGENCE	Name of finished Product	AI-Powered Podcast Analysis & Intelligent Segmentation System	
Name of Supervisor With e-mail id	Ms. Dikshita springboardmentor22342d@gmail.com			
Main points of the Week				
Studied fundamentals of Speech-to-Text (STT) and Automatic Speech Recognition (ASR). Explored open-source STT engines: • Whisper • Vosk • Coqui STT • Kaldi • SpeechBrain Studied transformer-based models and on-device Speech-to-Text solutions. Analyzed speech datasets and selected suitable dataset for experimentation. Performed transcript quality inspection and identified challenges including noise, accents, and overlapping speech. Implemented Faster Whisper for audio transcription. Generated and verified transcripts for multiple audio samples. Saved transcript outputs in structured text/CSV format.				

STUDENT'S WEEKLY DIARY

Week-2			DATE	22 Feb 2026
Dept./Division	ARTIFICIAL INTELLIGENCE	Name of finished Product	AI-Powered Podcast Analysis & Intelligent Segmentation System	
Name of Supervisor With e-mail id	Ms. Dikshita springboardmentor22342d@gmail.com			
Main points of the Week				
Implemented TextTiling baseline for podcast segmentation. Performed transcript sentence splitting while preserving timestamp alignment. Generated sentence embeddings using MiniLM-L6-v2. Computed cosine similarity between consecutive transcript sentences. Created similarity curve visualizations for identifying topic boundaries. Implemented threshold-based boundary detection for segmentation. Grouped sentences into meaningful topic segments. Stored segmentation outputs in JSON format with timestamps. Prepared evaluation scripts for Boundary Precision and Recall.				

Week-3			DATE	2 Mar 2026
Dept./Division	ARTIFICIAL INTELLIGENCE	Name of finished Product	AI-Powered Podcast Analysis & Intelligent Segmentation System	
Name of Supervisor With e-mail id	Ms. Dikshita springboardmentor22342d@gmail.com			
Main points of the Week				
Computed evaluation metrics for segmentation performance. Implemented dynamic threshold-based topic boundary detection. Generated similarity scores and graphical visualizations. Built SBERT-based segmentation pipeline for podcast episodes. Produced structured JSON output with timestamps and segment details. Compared predicted topic boundaries with manually annotated references. Created comparison table for: • Block size • k-value • Segment count • Segmentation observations Performed parameter tuning to optimize segmentation quality.				

Week-4			DATE	9 Mar 2026
Dept./Division	ARTIFICIAL INTELLIGENCE	Name of finished Product	AI-Powered Podcast Analysis & Intelligent Segmentation System	
Name of Supervisor With e-mail id	Ms. Dikshita springboardmentor22342d@gmail.com			
Main points of the day/Week				
Conducted parameter tuning using different block sizes and threshold values. Performed keyword extraction using KeyBERT. Generated segment-level summaries using lightweight summarization models. Integrated sentiment analysis into transcript segments. Computed sentiment polarity and numeric sentiment scores. Stored segment embeddings for semantic search functionality. Finalized structured JSON schema for transcript output. Validated timestamp alignment and segment consistency.				

<table>
<tr><td colspan="3">Week-5</td><td>DATE</td><td>16 Mar 2026</td></tr>
<tr><td>Dept./Division</td><td>ARTIFICIAL INTELLIGENCE</td><td>Name of finished Product</td><td colspan="2">AI-Powered Podcast Analysis & Intelligent Segmentation System</td></tr>
<tr><td>Name of Supervisor With e-mail id</td><td colspan="4">Ms. Dikshita
springboardmentor22342d@gmail.com</td></tr>
<tr><td colspan="5">Main points of the day/Week</td></tr>
<tr><td colspan="5">Implemented transcript navigation and segment indexing.
Enabled keyword-based and semantic search for podcast segments.
Designed and developed user interface for transcript visualization.
Added display for:
• Segment timestamps
• Segment summaries
• Keywords
Implemented sentiment visualization across podcast segments.
Generated keyword clouds and topic highlights.
Created timeline-based segment visualization.
Tested system performance on multiple podcast episodes.
Verified segmentation quality, keyword relevance, and summary accuracy.</td></tr>
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<table>
<tr><td colspan="3">Week-6</td><td>DATE</td><td>23 Mar 2026</td></tr>
<tr><td>Dept./Division</td><td>ARTIFICIAL INTELLIGENCE</td><td>Name of finished Product</td><td colspan="2">AI-Powered Podcast Analysis & Intelligent Segmentation System</td></tr>
<tr><td>Name of Supervisor With e-mail id</td><td colspan="4">Ms. Dikshita
springboardmentor22342d@gmail.com</td></tr>
<tr><td colspan="5">Main points of the day/Week</td></tr>
<tr><td colspan="5">Enhanced transcript navigation and metadata visualization.
Improved segment indexing and retrieval mechanism.
Added support for timestamp-based navigation between podcast segments.
Integrated segment summaries and keyword highlighting in interface.
Optimized backend processing for transcript loading and display.
Tested system responsiveness using multiple podcast episodes.
Verified proper synchronization between transcript segments and timestamps.
Improved overall usability of transcript exploration.</td></tr>
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Week-7		DATE	30 Mar 2026
Dept./Division	ARTIFICIAL INTELLIGENCE	Name of finished Product	AI-Powered Podcast Analysis & Intelligent Segmentation System
Name of Supervisor With e-mail id	Ms. Dikshita springboardmentor22342d@gmail.com		
Main points of the day/Week			
Implemented sentiment trend visualization across podcast timeline. Generated graphical representation of emotional tone shifts. Improved dominant topic extraction using keyword analysis. Enhanced segment summaries for better readability and interpretation. Performed testing on multiple podcast episodes for consistency evaluation. Verified sentiment realism and summary accuracy. Improved relevance of extracted keywords across segments. Validated logical coherence of generated topic boundaries.			

Week-8		DATE	06 Apr 2026
Dept./Division	ARTIFICIAL INTELLIGENCE	Name of finished Product	AI-Powered Podcast Analysis & Intelligent Segmentation System
Name of Supervisor With e-mail id	Ms. Dikshita springboardmentor22342d@gmail.com		
Main points of the day/Week			
Performed end-to-end testing of the complete podcast analysis pipeline. Validated Speech-to-Text transcription quality using Faster Whisper. Verified segmentation quality using SBERT-based topic boundary detection. Tested semantic search and keyword-based retrieval system. Improved interface visualization for transcript segments and analytics. Verified sentiment analysis outputs and emotional trend visualizations. Conducted final testing for timestamp accuracy and segment consistency. Prepared complete project documentation and workflow explanation for final submission.			

K Dikshita

Signature of Industry Supervisor